The depth gain does not transport: the ~2M floor stands

N=48, degree 5, 4 colours, leak-free | the L=12 width ladder repeated at L=16 | K 51668331, 6 cells, 3 seeds | one knob from J2

THE STANDING MINIMUM DOES NOT MOVE, AND THAT IS THE RESULT

KA (attention-only L=16 d160, 1,759,738 trainable) came back 0.757 on 3/3 seeds crossing, and KB (1,145,658) 0.766 on 3/3. Neither satisfies the registered rule (3 of 3 AND median at or below 0.74), so attention-only L=16 d192 at 2,504,890 / 0.580 remains the standing minimum. Stage K was registered to be able to say this: the breaking point is the result, and a rung that fails the rule is reported at the same volume as one that passes.

WHY THIS WAS WORTH SIX CELLS

Every rung below 2M on record was an L=12 rung, so 'there is a floor near 2M parameters on this task' was really a statement about one depth. At a fixed d192 the step L=12 -> L=16 moved rho_50 0.754 -> 0.580, a gain of 0.174 = 1.17 s*, and carried the rung across 0.74. K0 asked whether that same step carries d160 over too: 0.773 minus 0.174 = 0.599, registered as [0.55, 0.7] with the robust call being 'KA IS FREE on 3/3'. Observed 0.757.

KB (L=16 d128, 1,145,658) -- CROSSED BUT NOT FREE, falsifying K1

KB crosses 3/3 at rho_50 0.766 where d128 CENSORED at L=12, so depth does pay for a per-layer width bottleneck and the CENSORING BOUNDARY -- the narrowest rung this eval can read at all -- moves to ~1.15M. That is not the free line: 0.766 is +0.18 s* past 0.74, so K2 does not fire. 2/3 seeds left the 1.64 plateau.

WHAT IS NOT CLAIMED

The best solver on this eval at any size is still the COMPLETE model at L=12, rho_50 0.385 at 9,638,458 trainable; 'minimal' has never meant 'best'. Every number here is N=48 -- the frozen bar s* = 0.1484, finding 44's 1.64 plateau and the 0.33 grid floor are all N=48 quantities and none of them is reused at another problem size. Whether any of this survives N=96 is stage F (51656823), running separately.

rho_50 = realized hint density at which solve rate crosses 0.5 on a 14-point x 400-instance grid, LOWER IS BETTER, reported CENSORED when no bin crosses. mean_solve = unweighted mean over the 14 bins, which never censors. s* = 0.1484 / 0.2918 was frozen before any of these cells ran and the free line 0.74 is the hub plus one s*. Commit 5a1f4f6.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-57	`N0 N1 ... N47`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`

`SOLVED`

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	`BEGIN_HINTS N2 COLOR_0 N5 COLOR_1 N6 COLOR_0 N10 COLOR_0 N12 COLOR_3 N13 COLOR_2 ... (39 tokens)`
GRAPH	`BEGIN_GRAPH N0 N23 N15 N26 N4 N31 N1 N29 N5 N7 N6 N22 ... (289 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N1 COLOR_2 N9 COLOR_1 N20 COLOR_3 N23 COLOR_0 N24 COLOR_3 N33 ... (73 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 29 deductions,
58 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000004, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

19 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N2=COLOR_0 N5=COLOR_1 N6=COLOR_0 N10=COLOR_0 N12=COLOR_3 N13=COLOR_2 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND          N1 COLOR_2 N9 COLOR_1 N20 COLOR_3 N23 COLOR_0 ...
  10 deduced
BEGIN_ROUND          N0 COLOR_1 N3 COLOR_2 N11 COLOR_1 N25 COLOR_1 ...
  6 deduced
BEGIN_ROUND          N19 COLOR_0
  1 deduced
BEGIN_ROUND          N34 COLOR_3
  1 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N1, neighbours N29 N5 N7 N6 N22

their colours known before this round:

```
[0, 1, 3]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N7 N22
```

only colour left: 2

```
so the trace emits  N1 COLOR_2
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

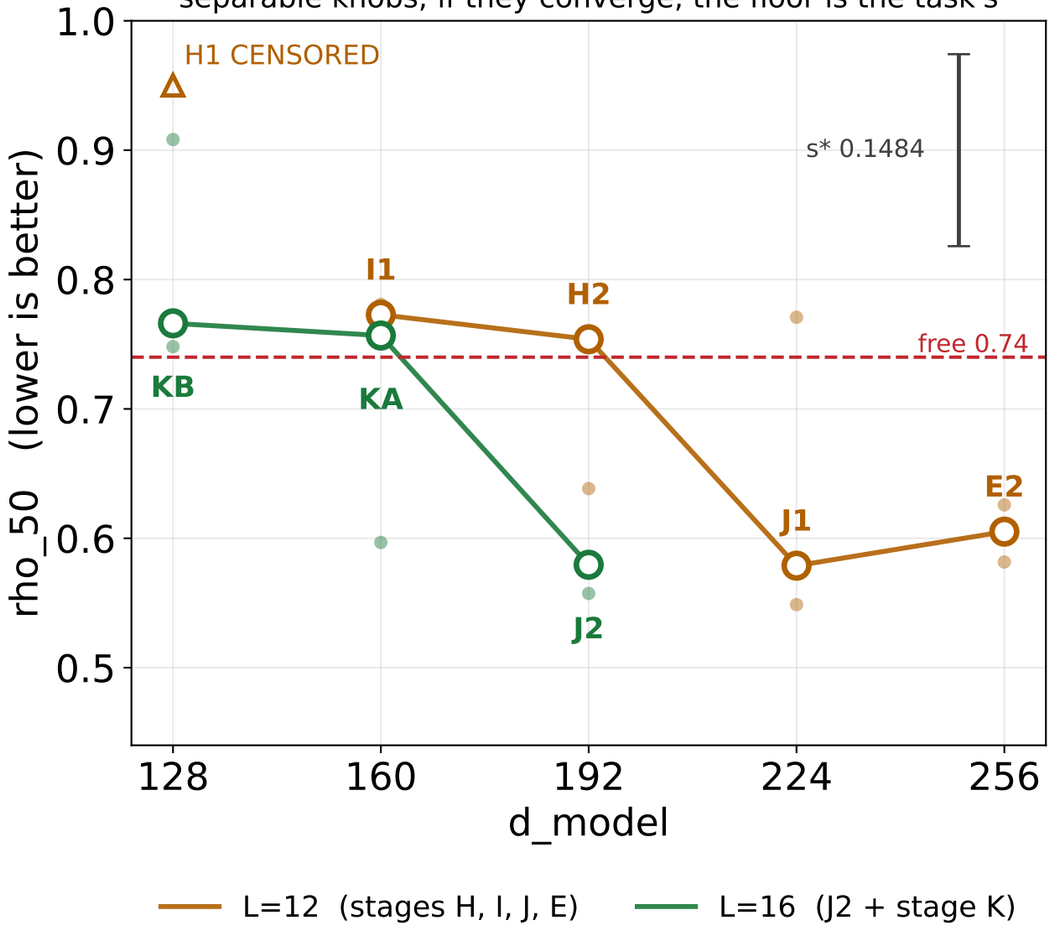
N=48, degree 5, 4 colours, --shuffle-adj --fall: the task every cell in this deck is trained and scored on. Stage K changes nothing about it -- only d_model moves, at a fixed L=16, so the two K rungs differ from the standing minimum by one knob.

T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

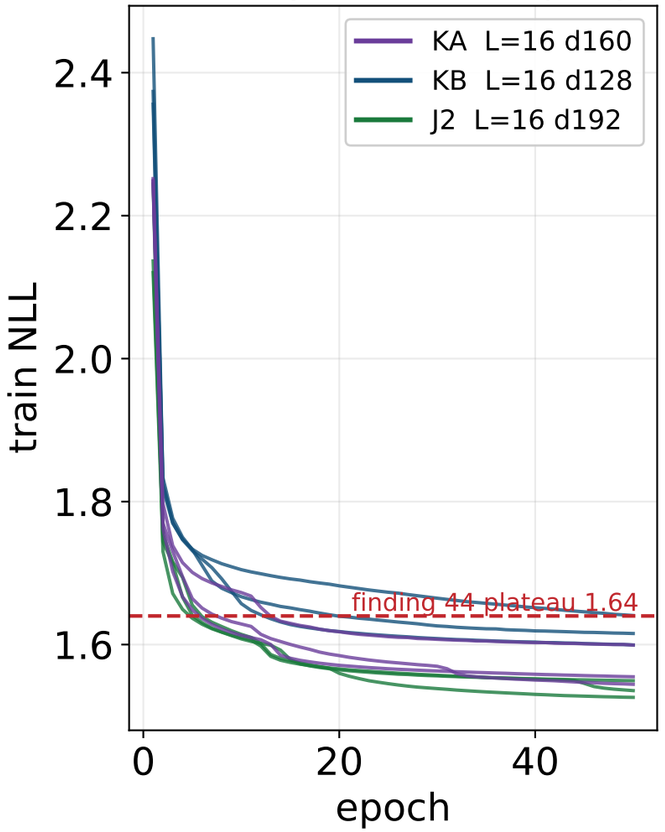
The same width ladder, run at two depths

Left: rho_50 against d_model, one curve per depth. Right: training, where a censored rung is read as capacity or as trainability

If the two curves are parallel, depth and width are separable knobs; if they converge, the floor is the task's



A rung that censors WITHOUT leaving the plateau is a trainability failure, not a capacity one



L=12 width ladder

d	trainable	rho_50	cross	esc	
128	879,418	CENSORED	1/3	1/3	H1
160	1,345,018	0.773	3/3	3/3	I1
192	1,908,922	0.754	3/3	3/3	H2
224	2,571,130	0.579	3/3	3/3	J1
256	3,331,642	0.605	3/3	3/3	E2

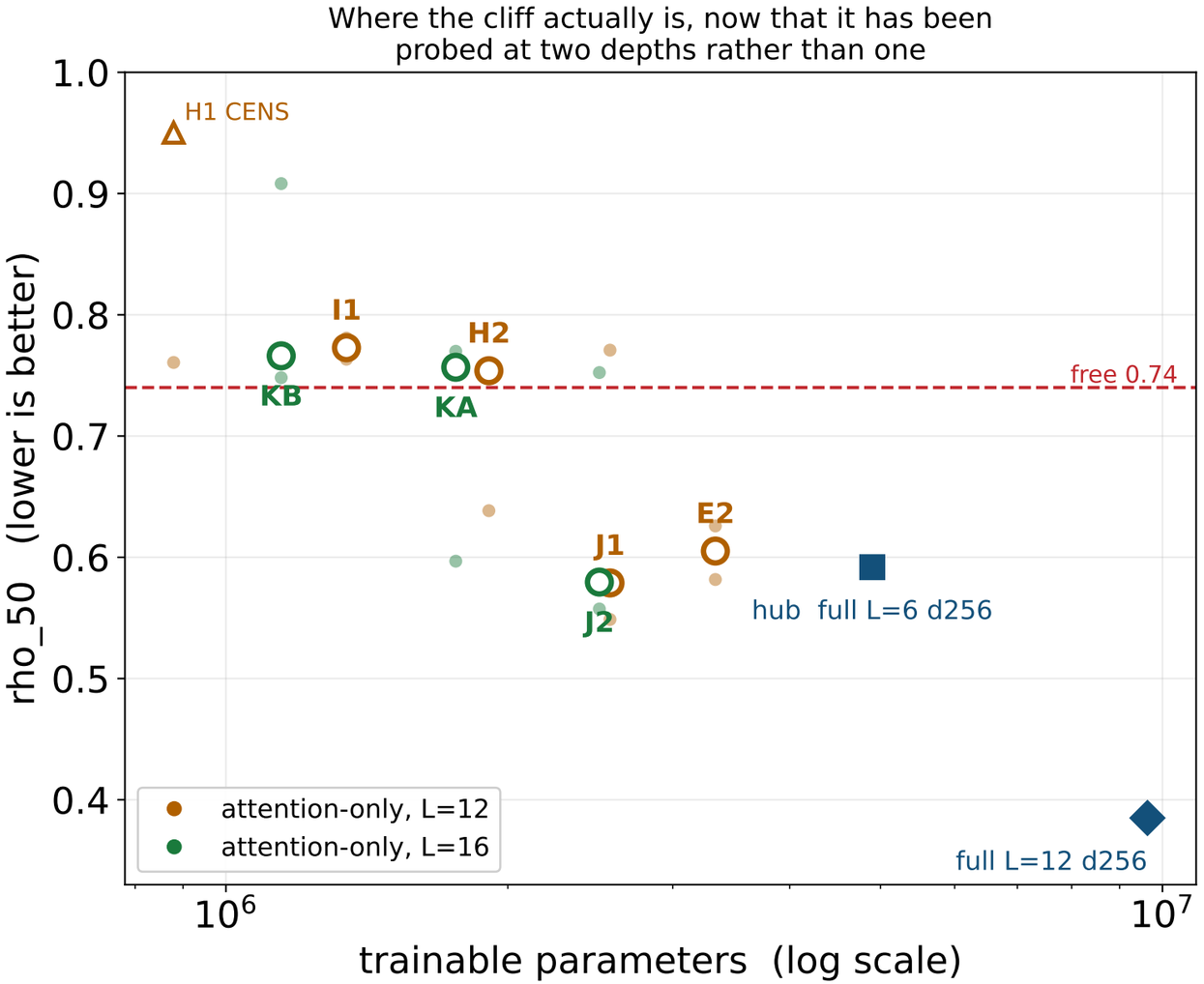
L=16 width ladder

d	trainable	rho_50	cross	esc	
128	1,145,658	0.766	3/3	2/3	KB
160	1,759,738	0.757	3/3	3/3	KA
192	2,504,890	0.580	3/3	3/3	J2

Open triangles are censored rungs, drawn at the grid edge 0.95 because that is all the eval knows about them; faint dots are individual seeds and the ringed marker is the 3-seed median, which is the quantity every verdict is read on. All cells attention-only, 240k instances x 50 epochs, batch 32, constant lr 1e-3, no warmup, --shuffle-adj --fall. KA 1,759,738 and KB 1,145,658 both matched their registered predictions (1,759,738 / 1,145,658) exactly. Commit 5a1f4f6.

The size-quality frontier, with stage K's two rungs added

Every attention-only rung this programme has run at N=48, against the two complete-model anchors



WHAT THE SHAPE OF THIS CURVE NOW SAYS. Before stage K the cliff was a one-depth measurement: everything below 2M was L=12, and the honest reading was 'a floor near 2M parameters on this task, and everything above it is flat'. Stage K is the test of that word 'task'.

The smallest rung that satisfies the registered rule anywhere on this plot is still J2 at 2,504,890, rho_50 0.580 -- the same rung as before stage K, so THE CLIFF HAS NOT MOVED. What did move is a lower line: the narrowest READABLE rung is now KB at 1,145,658. Readable is not free, and the rule reads the free line. Nothing here says a smaller model is a better one: the complete L=12 d256 model at 9,638,458 still solves at 0.385, far below anything on the attention-only curve.

The hub is the architecture s* is anchored to (full L=6 d256, rho_50 0.592 at 4,899,898); the free line is the hub plus one s*. Commit 5a1f4f6.

Every stage K prediction, scored against what was registered

stagek_score.py was committed (262b581) before any K cell had an eval on disk -- no verdict chosen after seeing a number

K0 FALSIFIED HIGH (registered: median in [0.55, 0.7], point estimate 0.599, robust call KA IS FREE on 3/3)

Observed 0.757 on 3/3 crossing and 3/3 plateau escapes, at 1,759,738 trainable. The point estimate was 0.773 (I1, the same width at L=12) minus the measured depth step 0.174 (H2 -> J2 at a fixed d192) = 0.599. The transport model does not survive: the depth gain measured at d192 does not carry to d160, so depth and width are not separable down here. LOW was registered as a falsification too, in our favour, earning no victory lap.

K1 FALSIFIED (registered: KB CENSORS, as H1 did at L=12)

KB crosses 3/3 at rho_50 0.766 where d128 CENSORED at L=12, so depth does pay for a per-layer width bottleneck and the CENSORING BOUNDARY -- the narrowest rung this eval can read at all -- moves to ~1.15M. That is not the free line: 0.766 is +0.18 s* past 0.74, so K2 does not fire. 2/3 seeds left the 1.64 plateau. K1 GATES THE ARCHITECTURE READING OF KB, as F1 gates F0: a censored rung that never left the plateau is a statement about optimization, not about what the architecture can represent.

K2 DID NOT FIRE -- the minimum stands | K3 before/after on the frozen bar

K2 is the standing-minimum rule unchanged from H3 = I3 = J3: 3 of 3 crossing AND median at or below 0.74, smaller wins the tie-break, and two of three is never enough (finding 51). BEFORE: attention-only L=16 d192, 2,504,890 trainable at rho_50 0.580. AFTER: unchanged. No rung in stage K satisfies the rule, so the before and after are the same number and the correct report is that the push failed at this depth.

K4 24/24 held-out rungs on the accuracy-to-solve curve, worst discriminating miss 0.0222 vs bar 0.0423

The curve is refit over the nine sufficiency dirs plus SCAN_K and asks whether stage K's rungs sit on it: 24 of 24 well-determined held-out rungs do, over 50 cells, 24 of them discriminating. Token accuracy remains a sufficient statistic over this grid. A rung that fell OFF this curve would be the interesting one -- it would mean stage K found a failure mode that token accuracy cannot see.

The LOO bar 0.0423 is the leave-one-out RMS of the depth ladder's own fit and was frozen with it. 'Discriminating' means the curve predicts solve > 0.10 for that cell, so landing on it is a real constraint rather than two numbers agreeing near zero. Commit 5a1f4f6.

Where this leaves the N=48 search

stage K probes the floor at a second depth | stage F (51656823) asks separately whether any of it transfers

WHAT STAGE K SETTLED

The ~2M floor is not an artefact of L=12. Repeating the width ladder one depth up did NOT produce a smaller free rung: the depth gain measured at d192 does not transport to narrower models, so the two knobs stop being separable below d192. Either way the useful output is the SHAPE of the two curves, which is now measured at two depths instead of one, and the breaking point is reported as the result rather than hidden behind the rungs that worked.

THE NEXT CUT, IF THERE IS ONE

There is no obvious next cut on this grid. The width floor held at two depths, which is the stronger version of the stage J claim, and the remaining open question is not architectural: it is whether the floor itself moves with problem size, which is stage F.

THE OPEN N=48 QUESTION STAGE J LEFT, STILL OPEN

J1's width curve is CONVEX on [192, 256]: d224 beat every straight line through its neighbours, so interpolating d_model between measured rungs systematically underestimates the in-between model. A fourth width point is the only way to know the shape of that stretch. It is deliberately NOT bundled into K, which moves depth, or into F, which moves problem size -- one knob at a time is the only reason any of these verdicts mean anything.

THE HONEST LIMITS OF EVERY NUMBER ON THESE SLIDES

All N=48, degree 5, 4 colours, one 14 x 400 rho grid ending at 0.33, one 240k x 50 epoch budget, three seeds per rung. Seed spread at fixed architecture exceeds the depth gaps the ladders are built from, which is why no single-seed rung is a measurement and why the median over three is the only quantity scored. s* = 0.1484 and the 1.64 plateau are N=48 constants and are not reused at N=96. The sufficiency result is correlational and is a statement about this grid's range.

Stage K: 6 cells, 3 seeds per rung, 51668331, --array=0-5%6 --time=10:00:00 -G 1 -p sganguli. Registered 11d42aa (sbatch) / 262b581 (scorer) before submission. Commit 5a1f4f6.